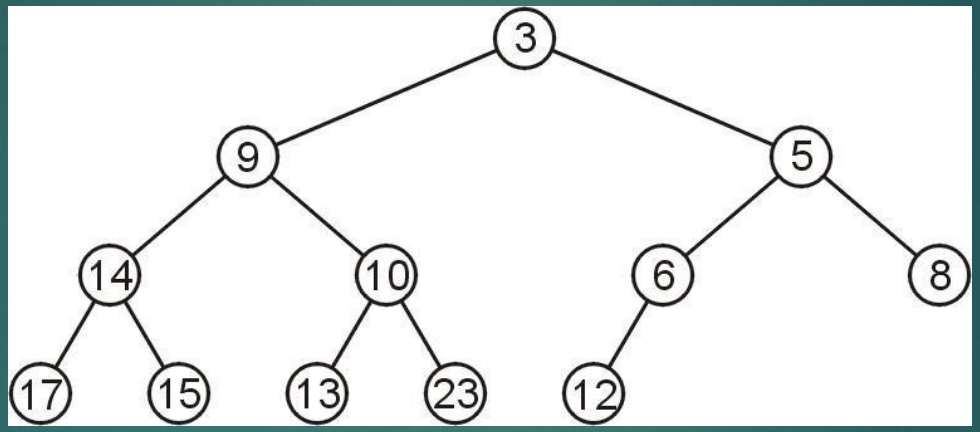
Week 13
Introduction to Heap

Array Storage

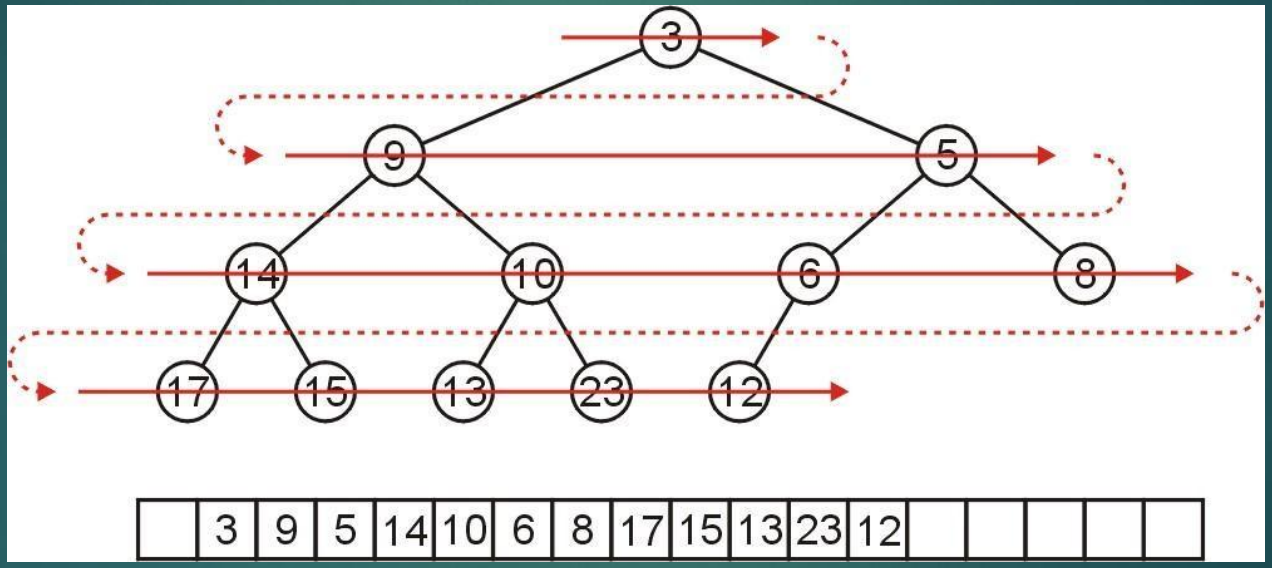
We are able to store a complete tree as an array

- Traverse the tree in breadth-first order, placing the entries into the array



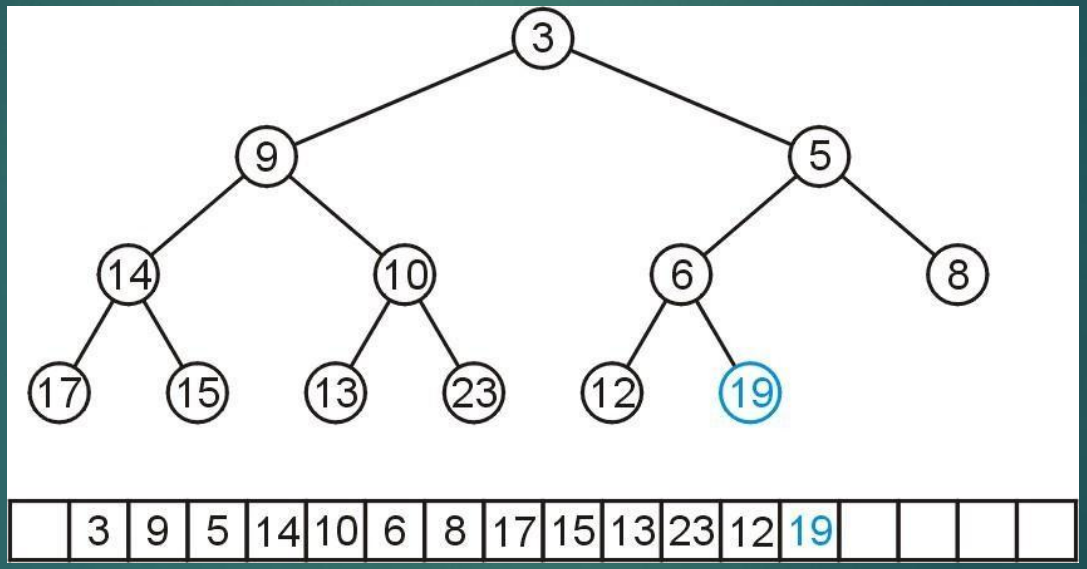
Array Storage (Insertion)

We can store this in an array after a quick traversal:



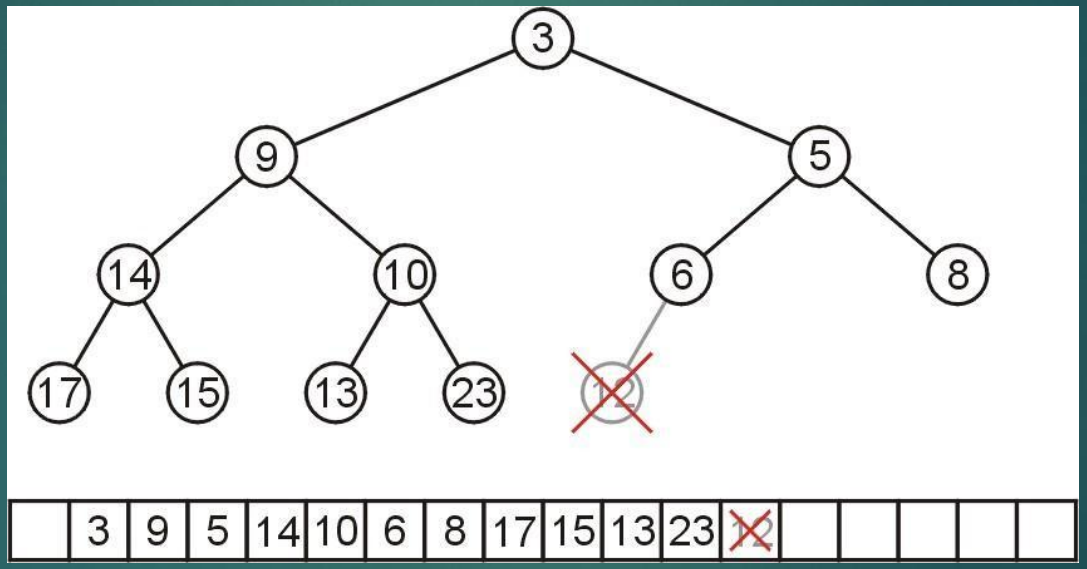
Array Storage (Insertion)

To insert another node while maintaining the complete-binary-tree structure, we must insert into the next array location



Array Storage (Deletion)

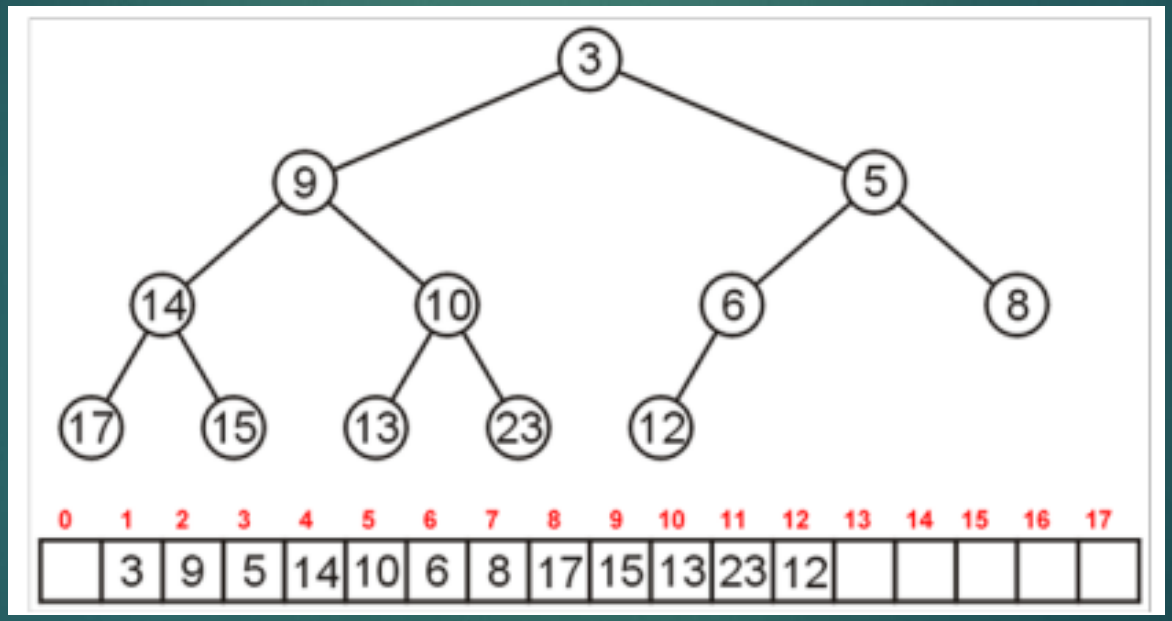
To remove a node while keeping the complete-tree structure, we must remove the last element in the array



Array Storage(Finding Parent and Child Nodes)

Leaving the first entry blank yields a bonus:

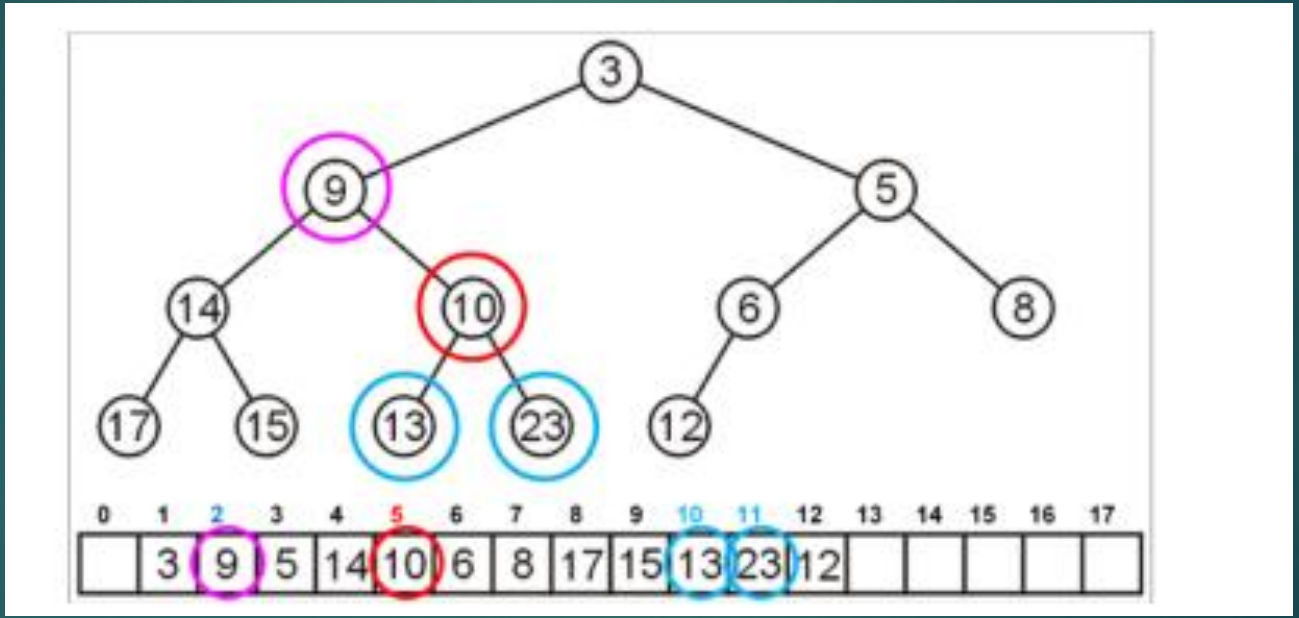
- The children of the node with index k are in $2k$ and $2k + 1$
- The parent of node with index k is in $k \div 2$



Array Storage

For example, node 10 has index 5:

– Its children 13 and 23 have indices 10 and 11, respectively



Outline

- Priority Queue
- Examples of Priority Queue
- Implementation details of Priority Queue
- Binary Heap
  - Min Heap
  - Max Heap
- Heap Sort

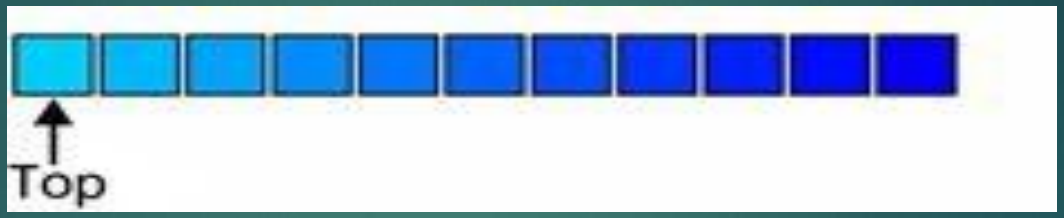
Priority Queue

With Queues

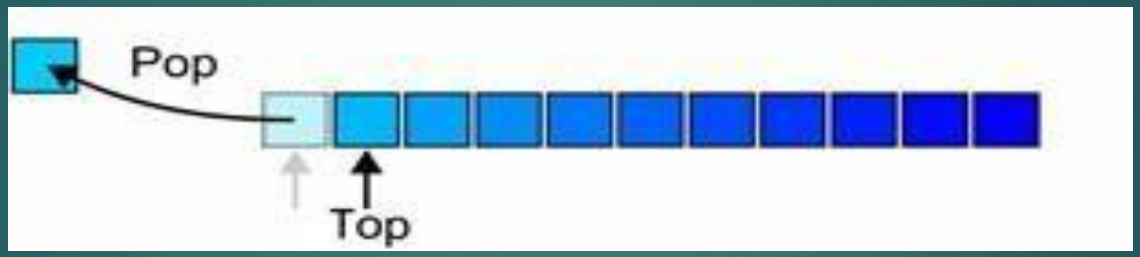
- The order may be summarized by *first in, first out*.
- If each object is associated with a priority, we may wish to pop that object which has highest priority.
- With each pushed object, we will associate a nonnegative integer (0, 1, 2, ...) where:
- The value 0 has the *highest* priority, and
- The higher the number, the lower the priority.

Operations

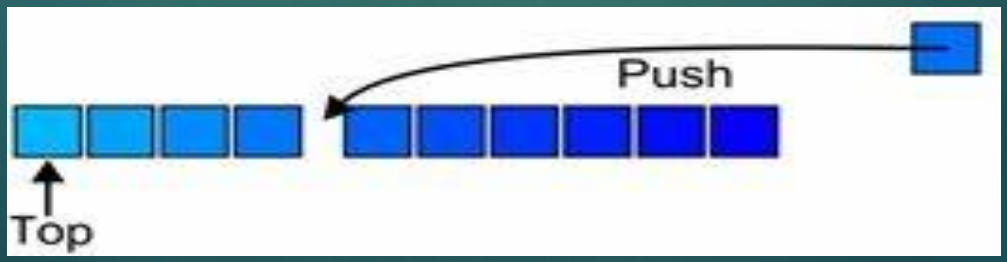
The top of a priority queue is the object with highest priority



Popping from a priority queue removes the current highest priority object:



Push places a new object into the appropriate place



Heap

- A **Heap** is a complete binary tree data structure that satisfies the heap property: for every node, the value of its children is greater than or equal to its own value.
- Heaps are usually used to implement priority queues, where the smallest (or largest) element is always at the root of the tree.

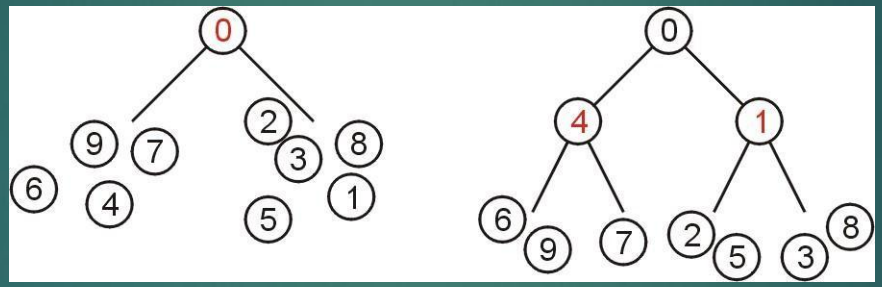
- Numerous other heaps exists:
  - D-ary heaps
  - Left-list heaps
  - Skew heaps
  - Binomial heaps
  - Fibonacci heaps
  - Bi-parental heaps

Heap

A non-empty binary tree is a min-heap if

The key associated with the root is less than or equal to the keys associated with either of the sub-trees (if any)

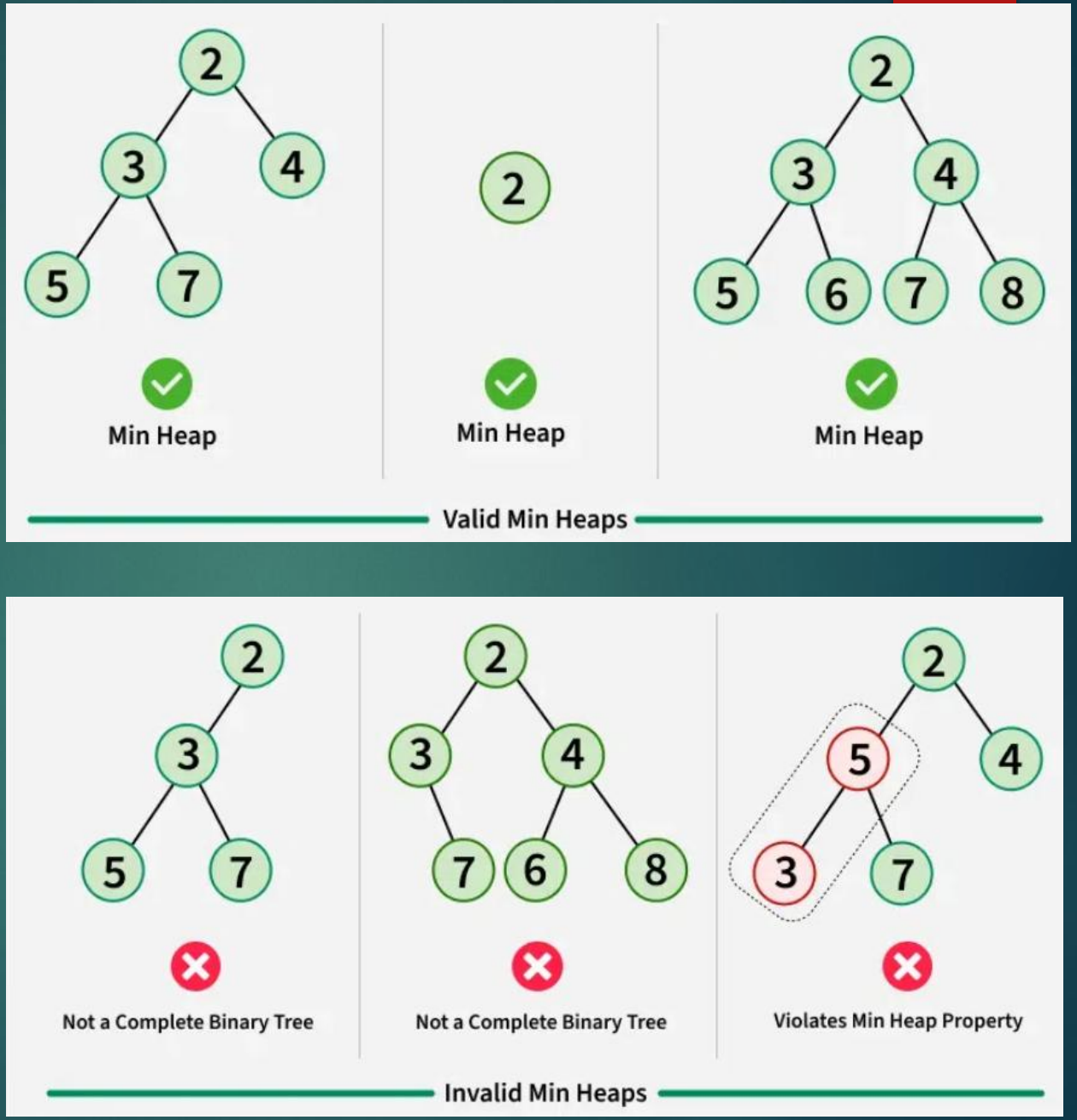
- Both of the sub-trees (if any) are also binary min-heaps



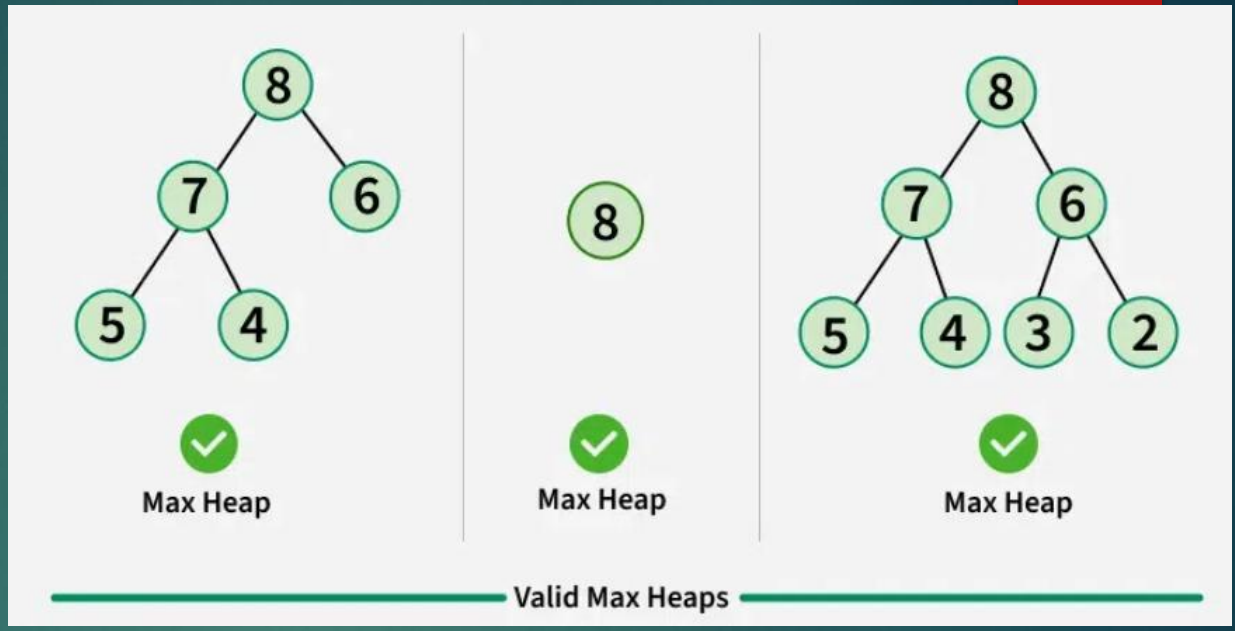
From this definition:

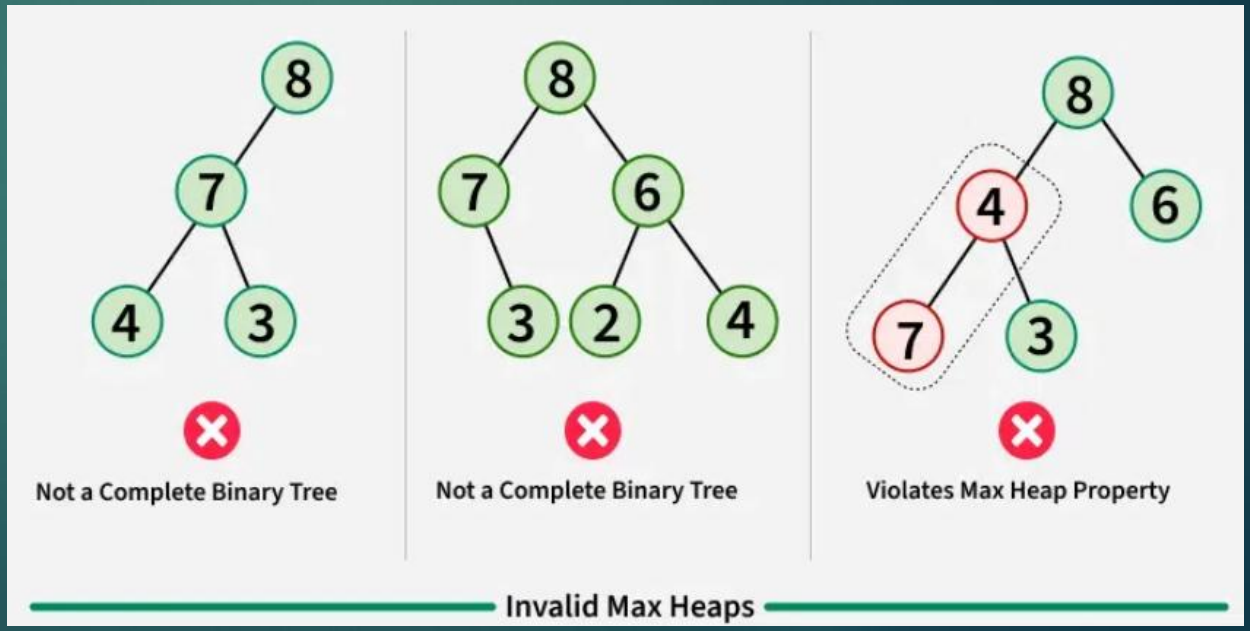
- A single node is a min-heap
- All keys in either sub-tree are greater than the root key

Heap


2
3
4
5
7
Min Heap
2
Min Heap
2
3
4
5
6
7
8
Min Heap
Valid Min Heaps
2
3
5
7
Not a Complete Binary Tree
2
3
4
7
6
8
Not a Complete Binary Tree
2
5
4
3
7
Violates Min Heap Property
Invalid Min Heaps

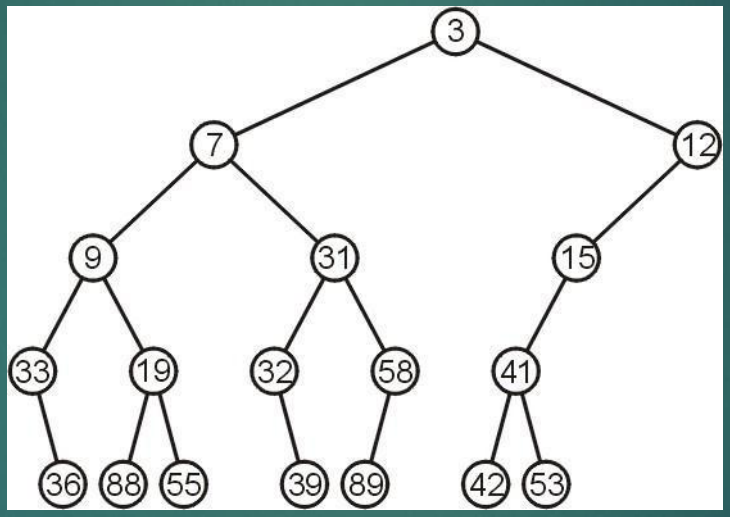
Heap


8
7
6
5
4
Max Heap
8
Max Heap
8
7
6
5
4
3
2
Max Heap
Valid Max Heaps


8
7
4
3
Not a Complete Binary Tree
8
7
6
3
2
4
Not a Complete Binary Tree
8
4
6
7
3
Violates Max Heap Property
Invalid Max Heaps

Example

This is a binary min-heap:



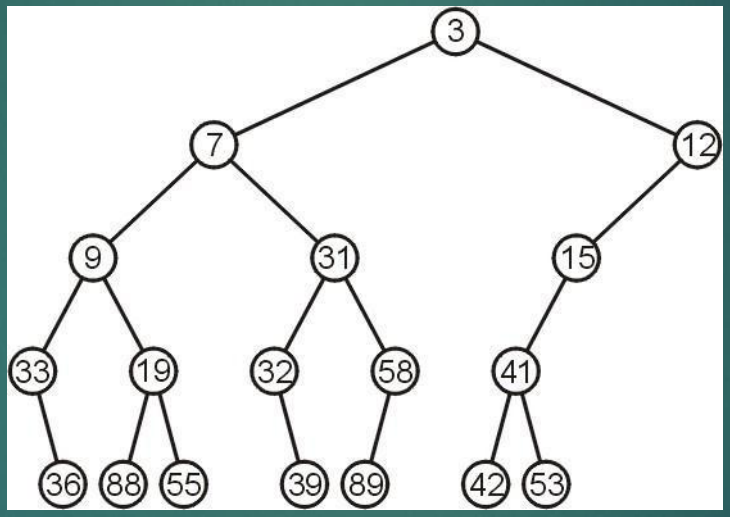
Operations on Heap

We will consider three operations:

- Top
- Pop
- Push

Example

We can find the top object in $\Theta(1)$ time: 3



Pop

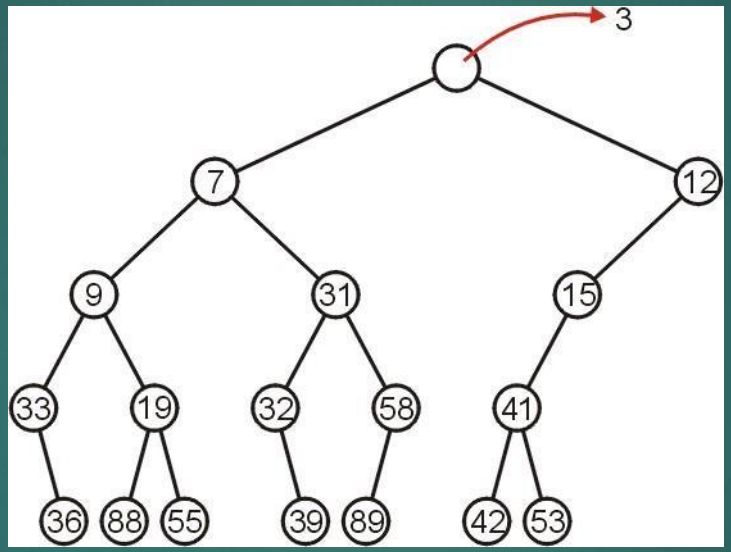
To remove the minimum object:

- Promote the node of the sub-tree which has the least value

  Recurs down the sub-tree from which we promoted the least value

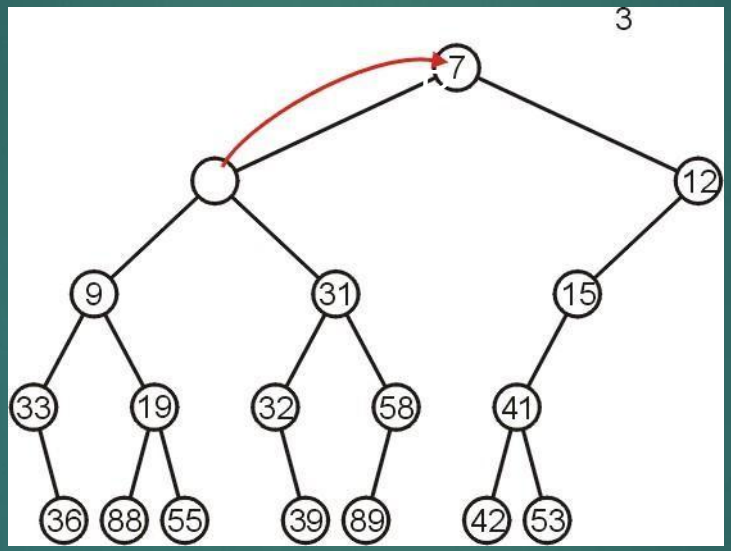
Pop

Using our example, we remove 3:



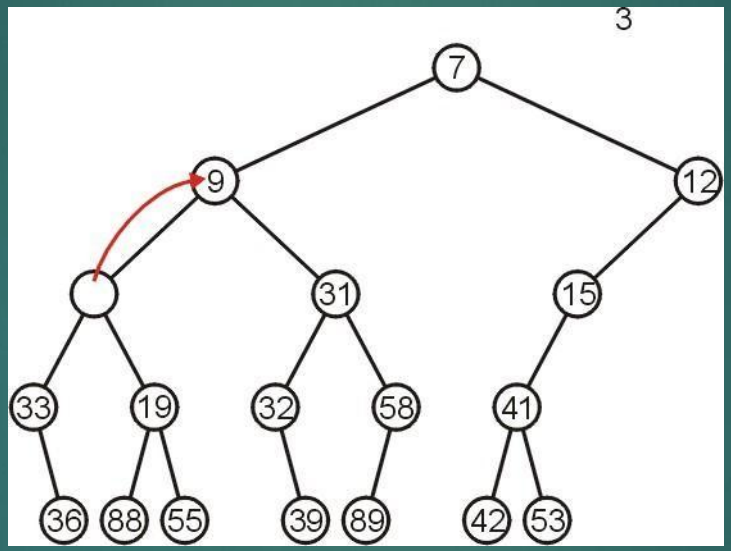
Pop

We promote 7 (the minimum of 7 and 12) to the root:



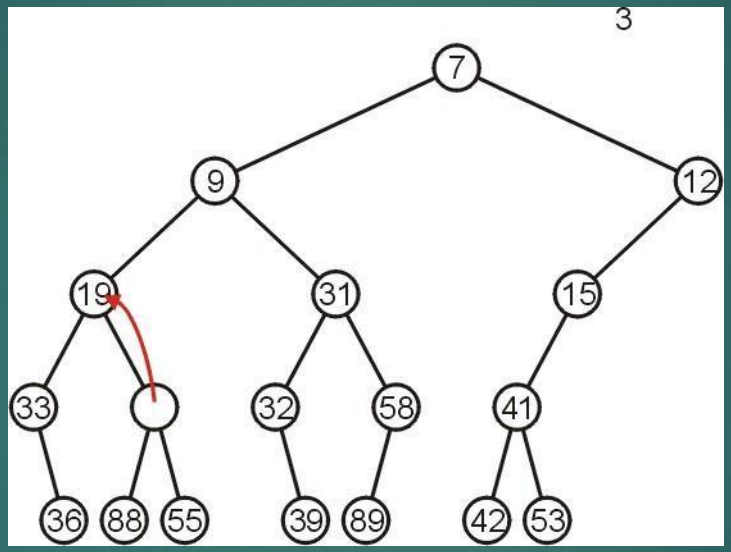
Pop

In the left sub-tree, we promote 9:



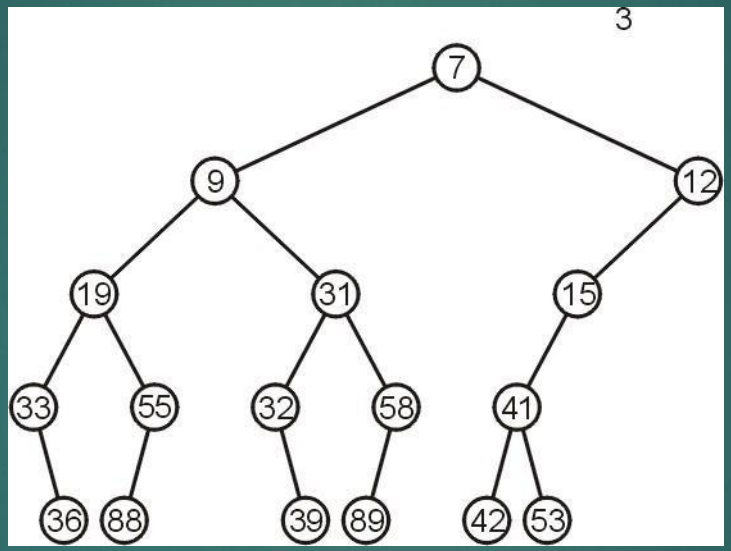
Pop

Recursively, we promote 19:



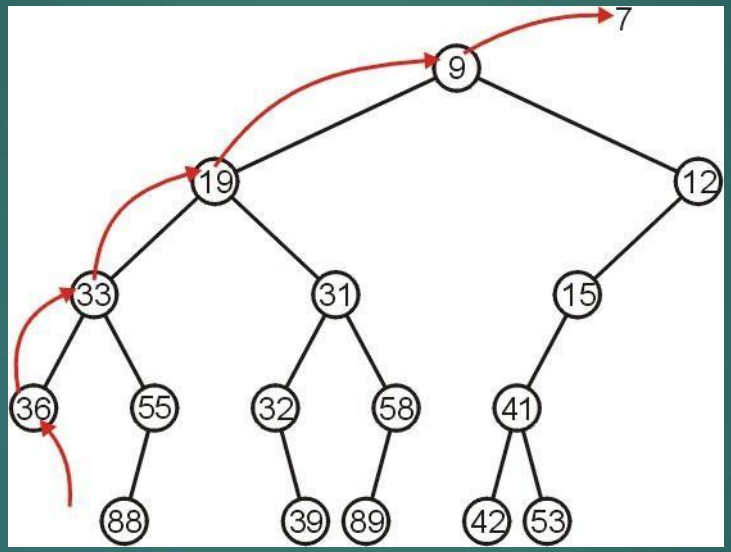
Pop

Finally, 55 is a leaf node, so we promote it and delete the leaf



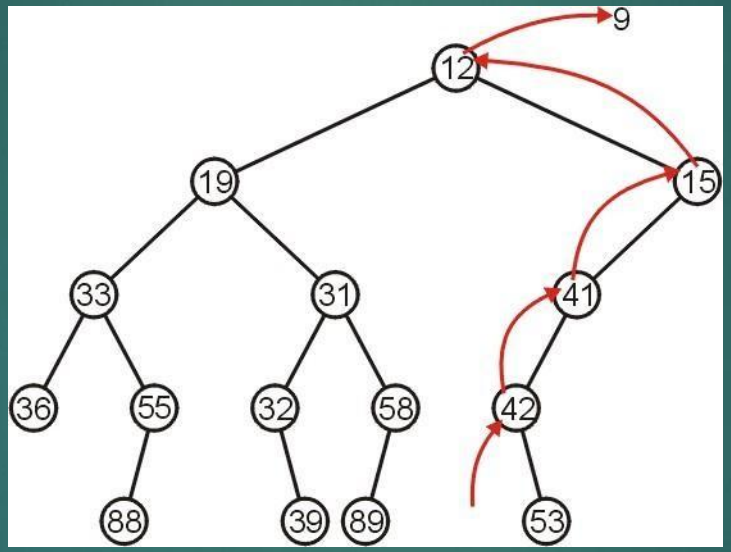
Pop

Repeating this operation again, we can remove 7:



Pop

If we remove 9, we must now promote from the right sub-tree:



Push

Inserting into a heap may be done either:

- At a leaf (move it up if it is smaller than the parent)
- At the root (insert the larger object into one of the subtrees)

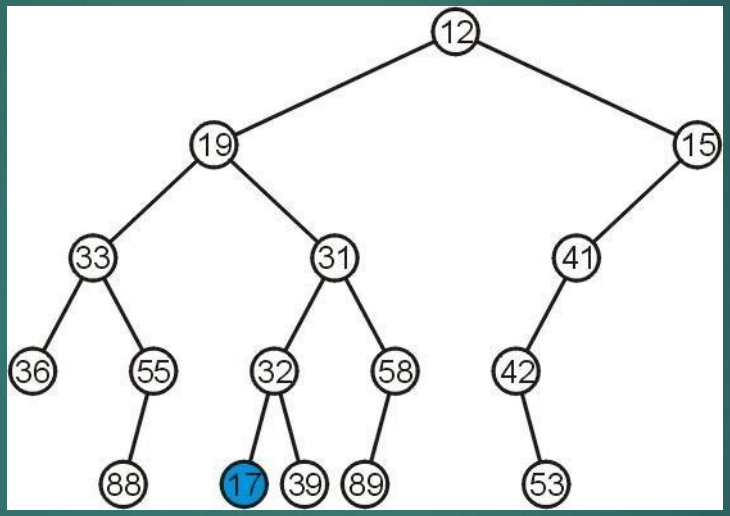
We will use the first approach with binary heaps

- Other heaps use the second

Push

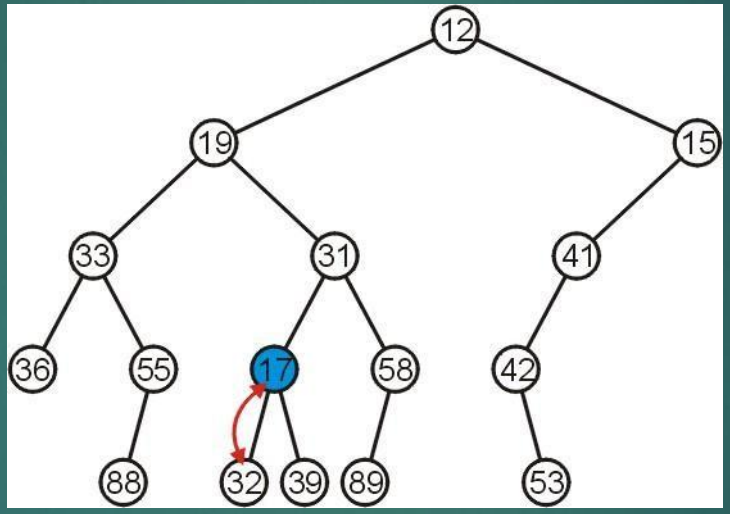
Inserting 17 into the last heap

– Select an arbitrary node to insert a new leaf node:



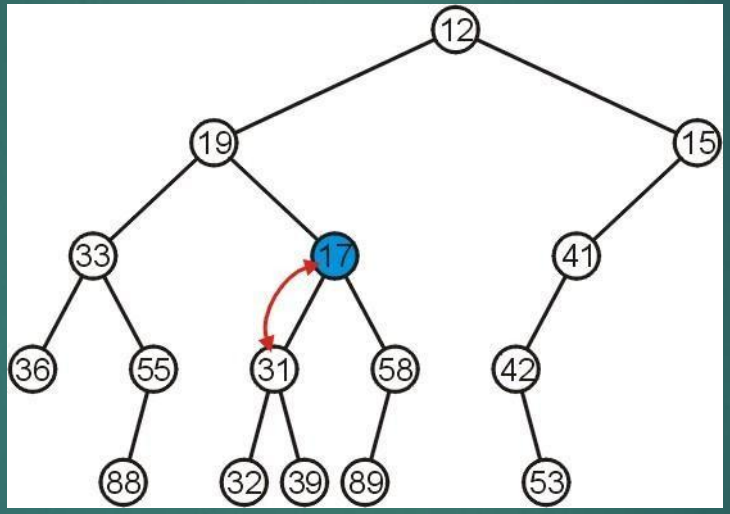
Push

The node 17 is less than the node 32, so we swap them



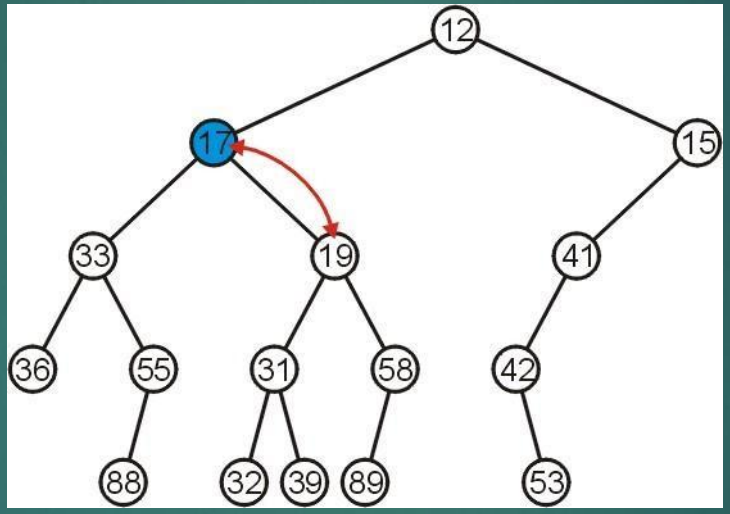
Push

The node 17 is less than the node 31; swap them



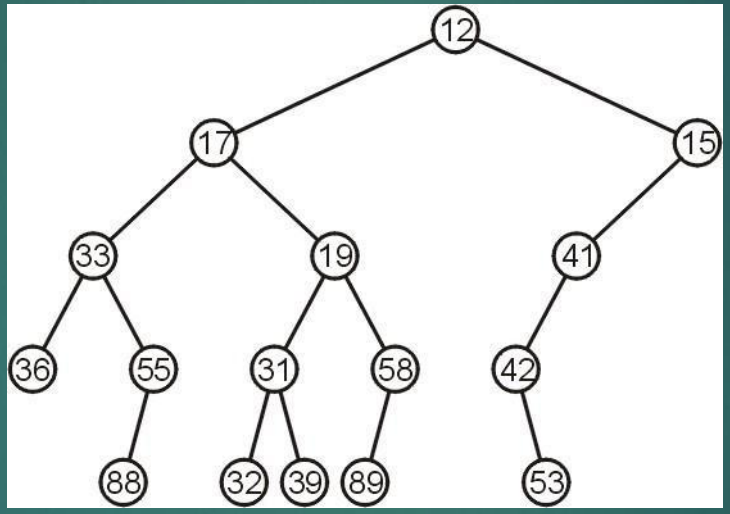
Push

The node 17 is less than the node 19; swap them



Push

The node 17 is greater than 12 so we are finished



Push

Observation: both the left and right subtrees of 19 were greater than 19, thus we are guaranteed that we don't have to send the new node down

This process is called *percolation*, that is, the lighter (smaller) objects move up from the bottom of the min-heap

Implementation Details

By using complete binary trees, we will be able to maintain, with minimal effort, the complete tree structure

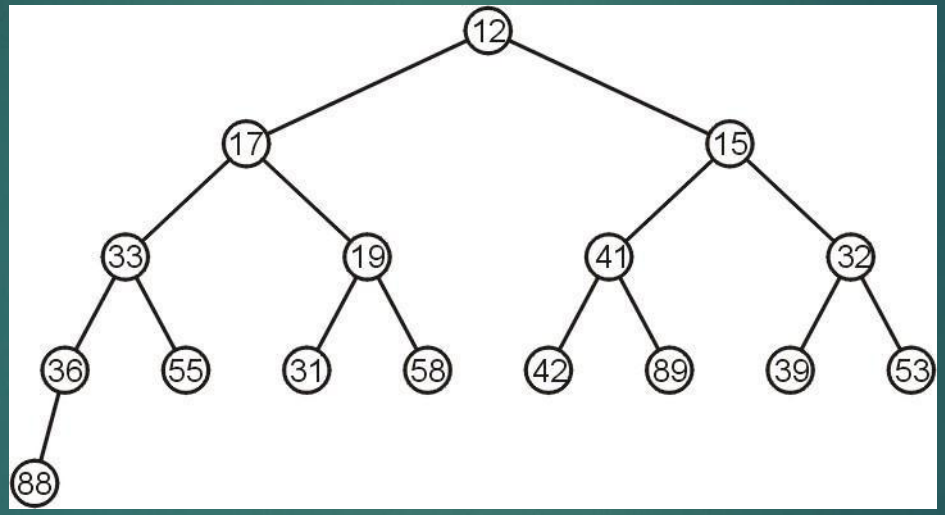
We have already seen

- It is easy to store a complete tree as an array

If we can store a heap of size n as an array of size $\Theta(n)$, this would be great!

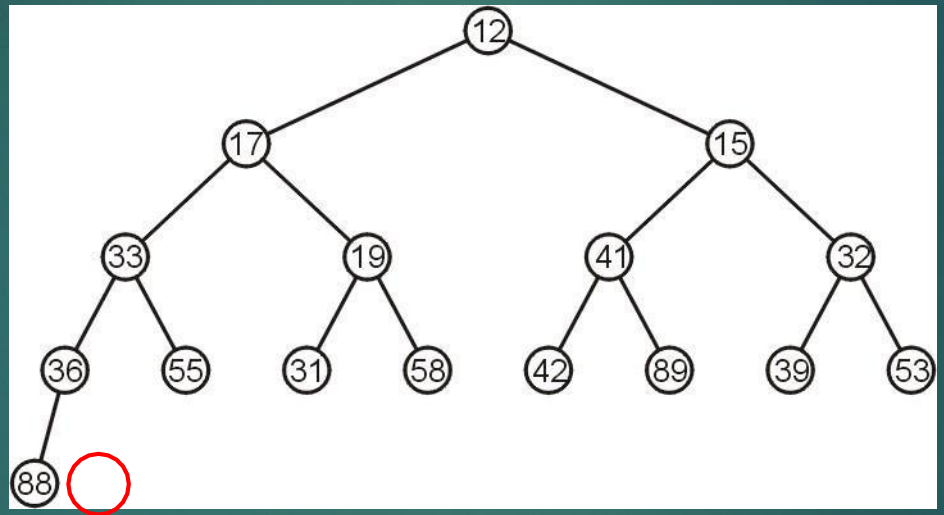
Example

For example, the previous heap may be represented as the following (non-unique!) complete tree:



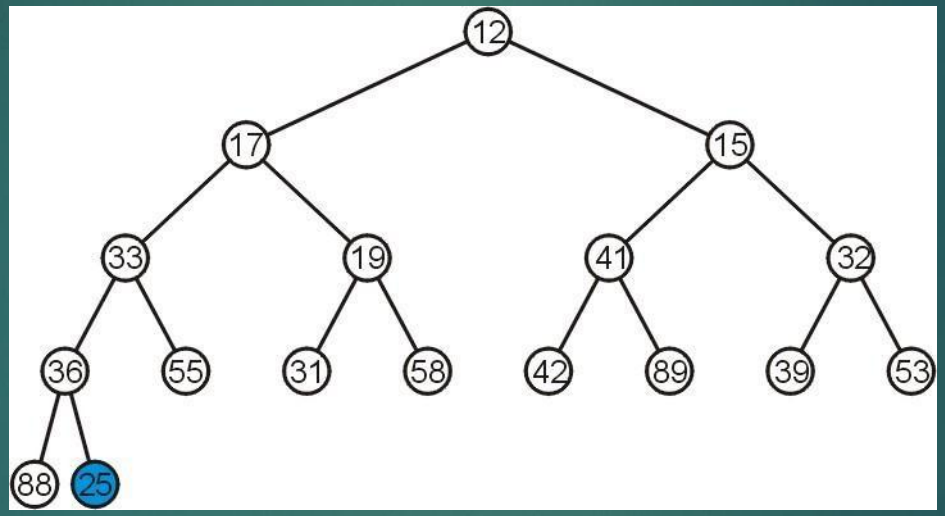
Complete Trees: Push

If we insert into a complete tree, we need only place the new node as a leaf node in the appropriate location and percolate up



Complete Trees: Push

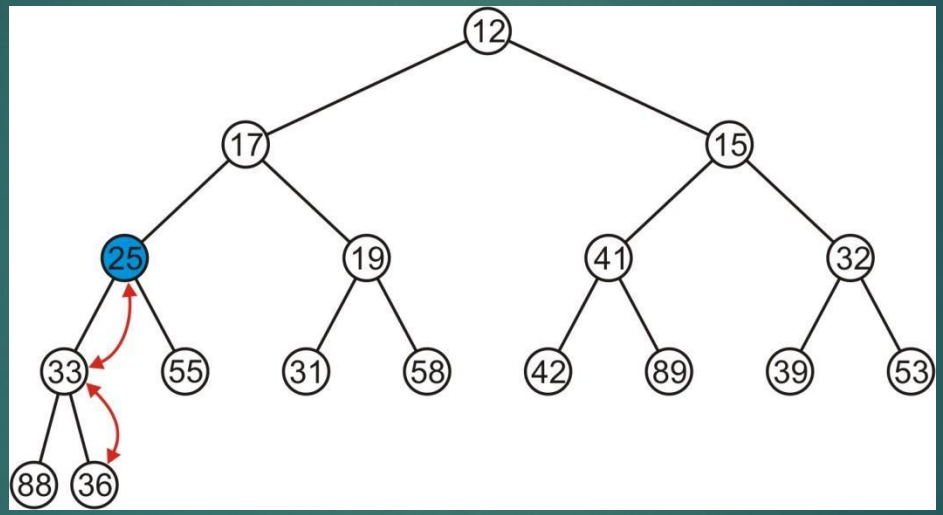
For example, push 25:



Complete Trees: Push

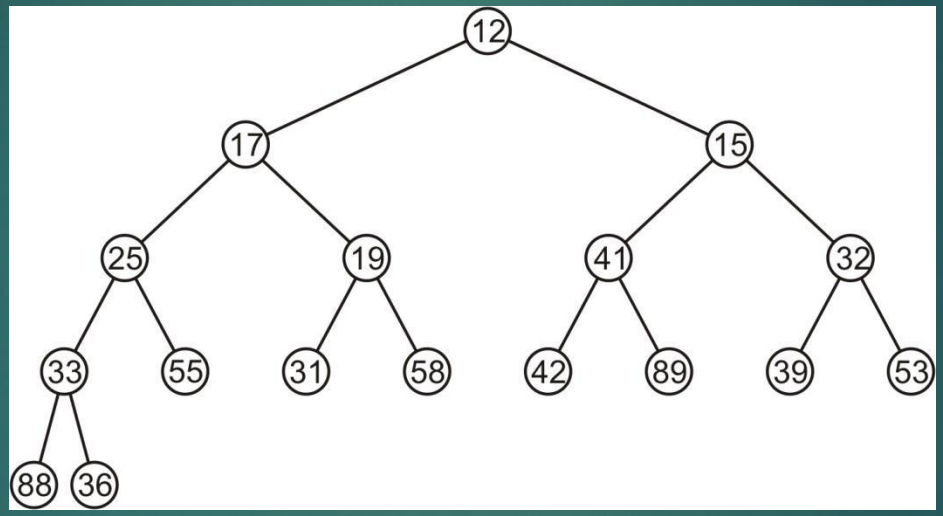
We have to percolate 25 up into its appropriate location

- The resulting heap is still a complete tree



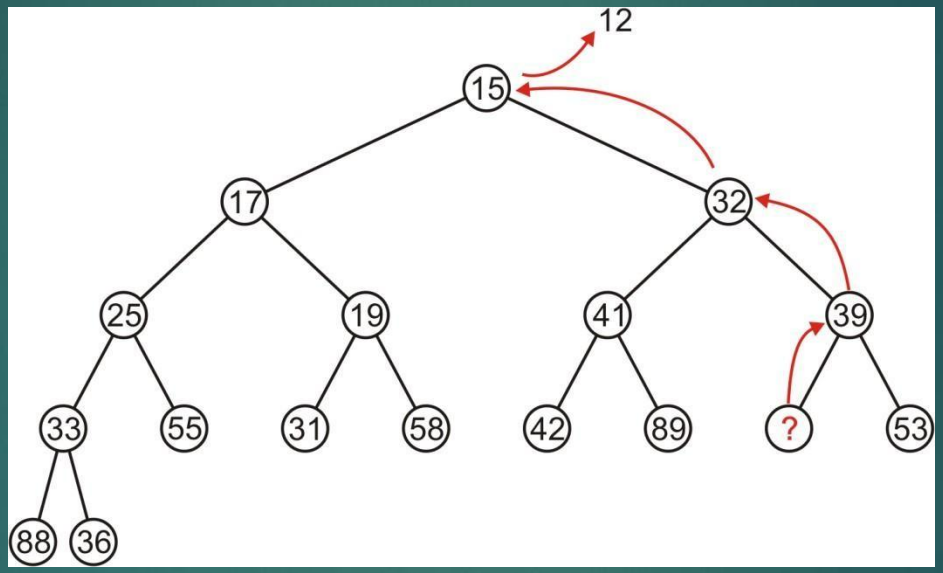
Complete Trees: Push

Suppose we want to pop the top entry: 12



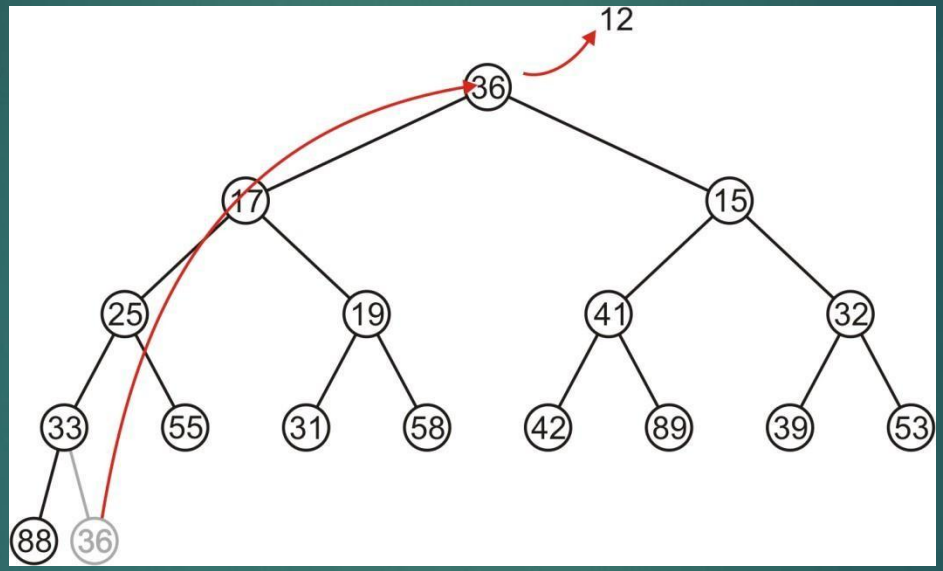
Complete Trees: Pop

Percolating up creates a hole leading to a non-complete tree



Complete Trees: Pop

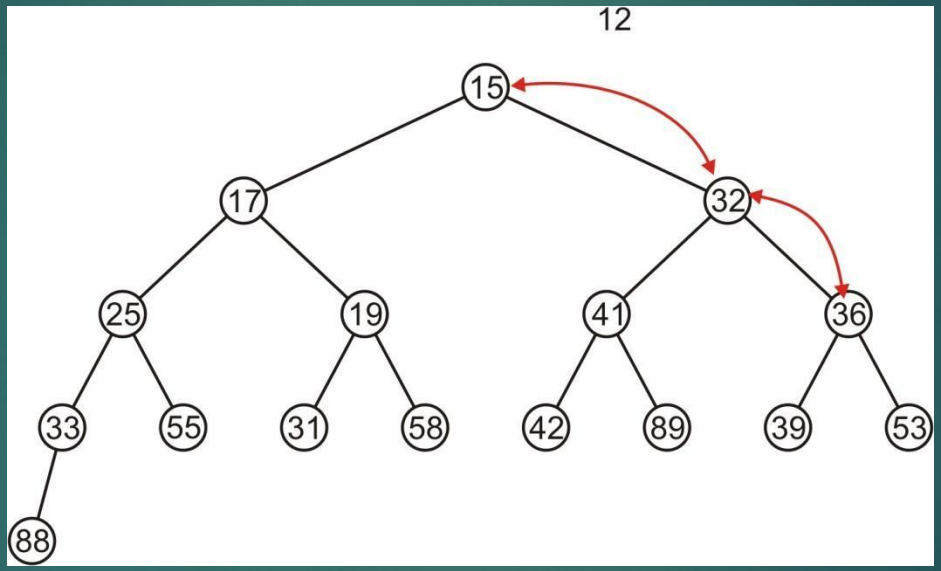
Alternatively, copy the last entry in the heap to the root



Complete Trees: Pop

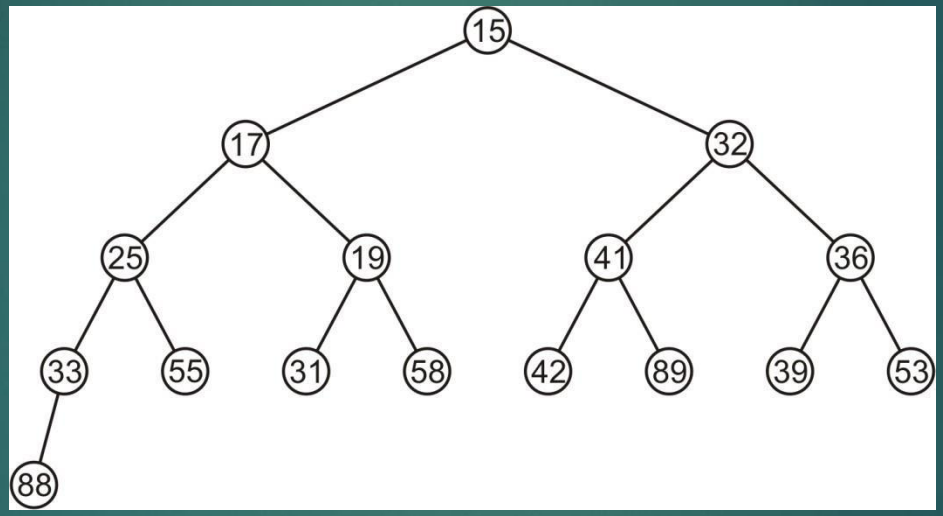
Now, percolate 36 down swapping it with the smallest of its children

- We halt when both children are larger



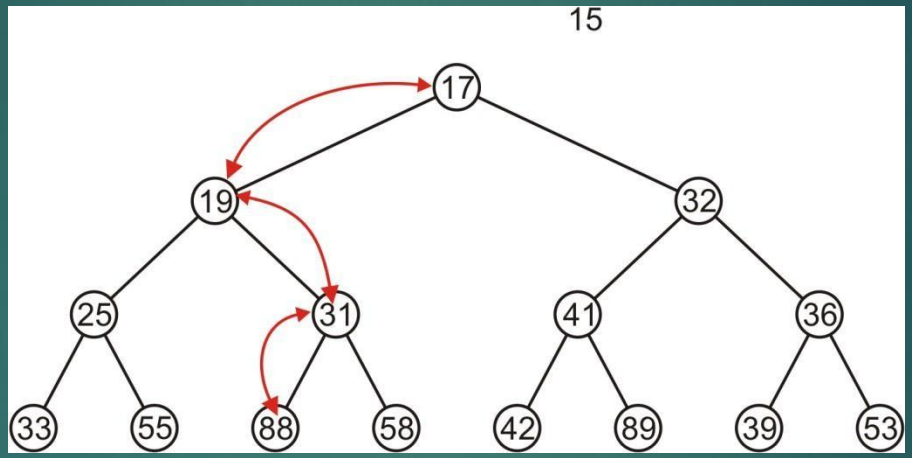
Complete Trees: Pop

The resulting tree is now still a complete tree:



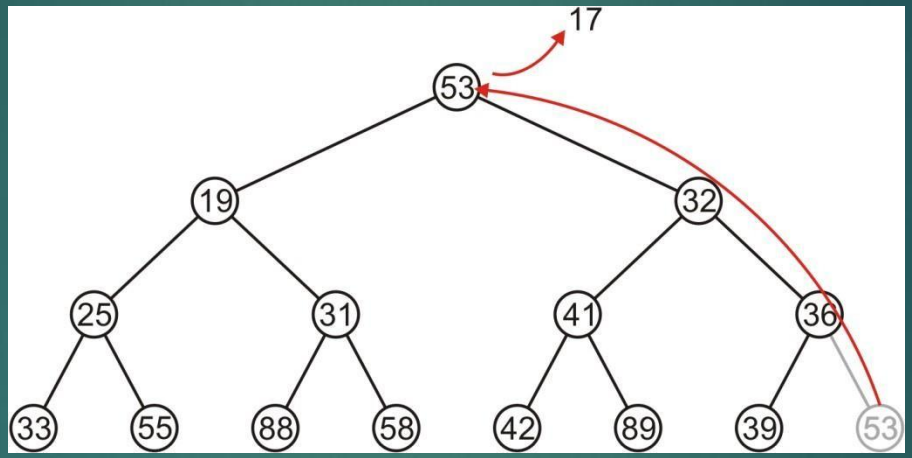
Complete Trees: Pop

This time, it gets percolated down to the point where it has no children



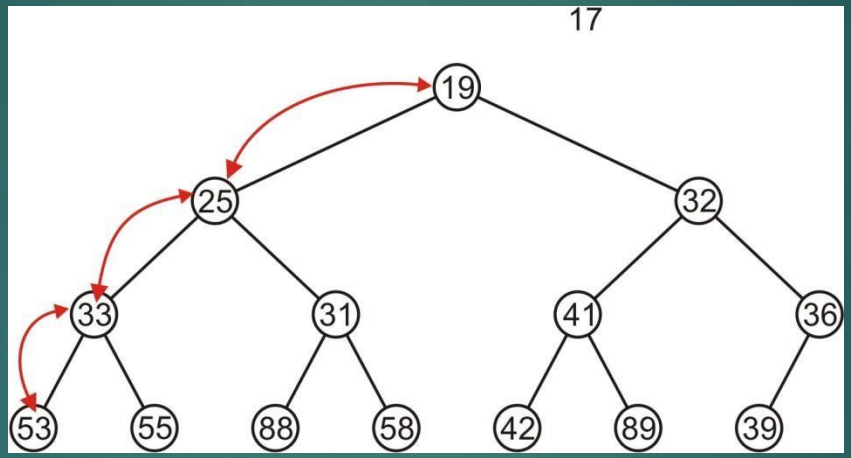
Complete Trees: Pop

In popping 17, 53 is moved to the top



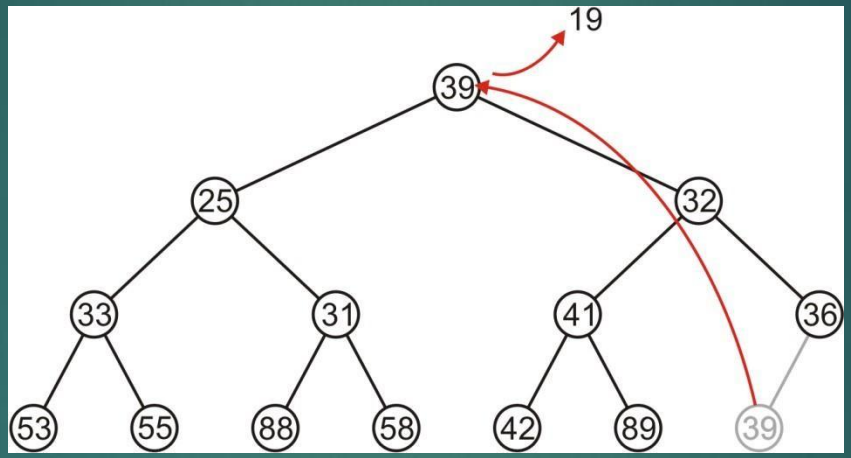
Complete Trees: Pop

And percolated down, again to the deepest level



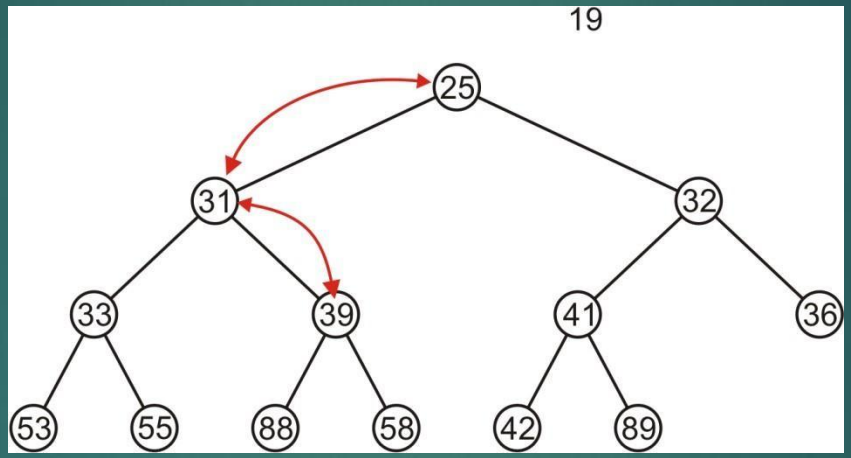
Complete Trees: Pop

Popping 19 copies up 39



Complete Trees: Pop

Which is then percolated down to the second deepest level



Complete Tree

- Therefore, we can maintain the complete-tree shape of a heap

- We may store a complete tree using an array:
  - A complete tree is filled in breadth-first traversal order
  - The array is filled using breadth-first traversal

- This structural requirement, along with the heap property, is essential for the heap data structure to guarantee **efficient O(log *n*) time complexity** for insertion and deletion operations, and for efficient memory representation using an array.
- Without the complete binary tree property, the tree could become unbalanced, degrading performance in the worst case to O(*n*) time, similar to a linked list.

Heap Sort

What is Heap Sort?
Heap Sort is a **comparison-based sorting algorithm** that uses a **binary heap** data structure.
It first builds a **max heap** and then repeatedly extracts the **maximum element** (root) to sort the array in **ascending order**.

Key Idea:
A **max heap** is a complete binary tree where each node is **greater than or equal to its children**.
Heap Sort leverages this property to sort elements efficiently.

Heap Sort

Algorithm Steps:

1.Build Max Heap

1. Convert the array into a max heap.
2. The largest element will be at the root (index 0).

2.Extract Max Element

1. Swap the root (largest element) with the **last element**.
2. Reduce the heap size by 1 (ignore the last element, as it's now sorted).

3.Heapify

1. Restore the max heap property for the remaining heap.
2. Repeat steps 2–3 until all elements are sorted.

Heap Sort

- Discussion

0	1	2	3	4	5	6	7	8	9
	3	1	4	7	8	6	0	2	5

- Step 01: Build Min Heap/Max Heap
- Step 02: Swap First index with Last index
- Step 03: Update heap size(HS)
- Step 04: Apply heapify on HS

Heap Sort

Example Dry Run

Let's sort the array:

`[4, 10, 3, 5, 1]`

Step 1: Build Max Heap

After heapify → `[10, 5, 3, 4, 1]`

Heap (level-wise):

```
        10
      /    \
     5      3
    / \
   4   1
```

Step 2: Extract max (10)

Swap root with last → `[1, 5, 3, 4, 10]`

Heapify `[1, 5, 3, 4]` → `[5, 4, 3, 1, 10]`

Step 3: Extract next max (5)

Swap → `[1, 4, 3, 5, 10]`

Heapify `[1, 4, 3]` → `[4, 1, 3, 5, 10]`

Heap Sort

Step 4: Extract next max (4)
Swap → `[3, 1, 4, 5, 10]`
Heapify `[3, 1]` → `[3, 1, 4, 5, 10]`
Step 5: Extract next max (3)
Swap → `[1, 3, 4, 5, 10]`
Final Sorted Array: `[1, 3, 4, 5, 10]`

Operation	Complexity
Building heap	O(n)
Each heapify	O(log n)
Overall sorting	O(n log n)
Space complexity	O(1) (in-place)

Heap Sort

Advantages
•Efficient for large datasets.
•In-place sorting (no extra memory).

Disadvantages
•Not stable (equal elements may change order).
•Slightly slower in practice than Quick Sort due to more swaps.

Run-time Analysis

Accessing the top object is $\Theta(1)$

Popping the top object is $\mathbf{O}(\ln(n))$

We copy something that is already in the lowest depth—it wil likely be moved back to the lowest depth

Priority Queues

Now, does using a heap ensure that that object in the heap which:

- has the highest priority, and
- of that highest priority, has been in the heap the longest

Consider inserting seven objects, all of the same priority (colour indicates order):

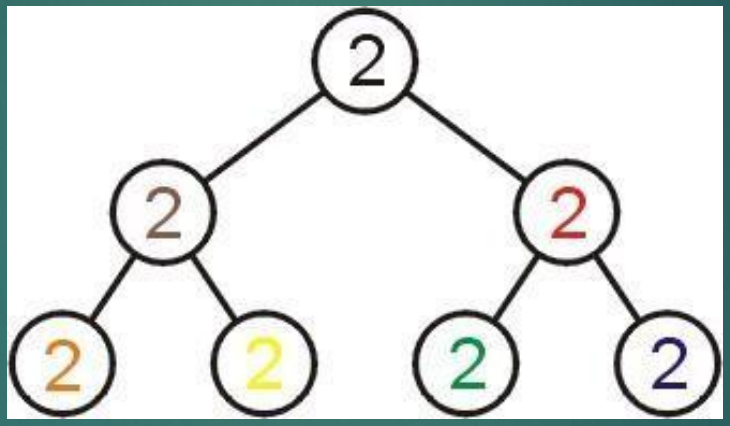
2, 2, 2, 2, 2, 2, 2

Priority Queues

Whatever algorithm we use for promoting must ensure that the first object remains in the root position

- Thus, we must use an insertion technique where we only percolate up if the priority is lower

The result:



Challenge:

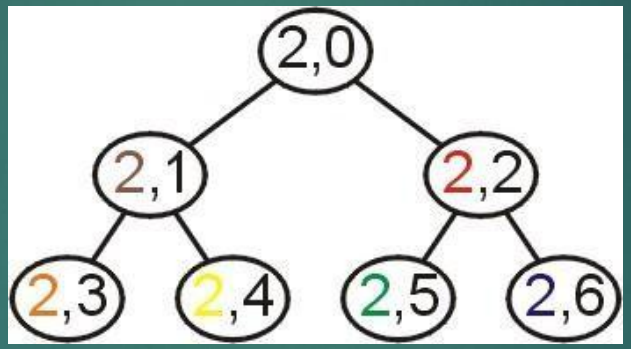
- Come up with an algorithm which removes all seven objects in the original order

Lexicographical Ordering

A better solution is to modify the priority:

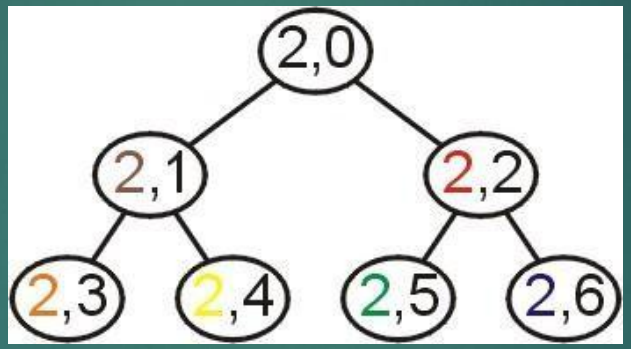
- Track the number of insertions with a counter k (initially 0)
- For each insertion with priority n, create a hybrid priority (n, k) where:

$(n_1, k_1) < (n_2, k_2)$ if $n_1 < n_2$ or ($n_1 = n_2$ and $k_1 < k_2$)



Priority Queues

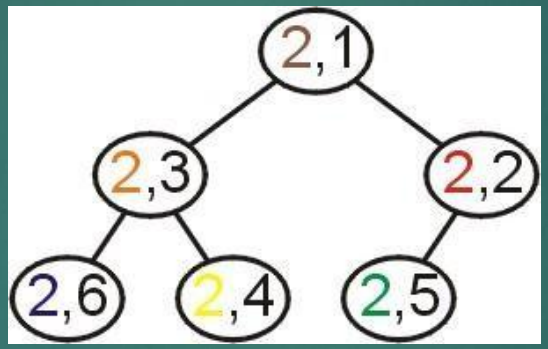
Removing the objects would be in the following order:



Priority Queues

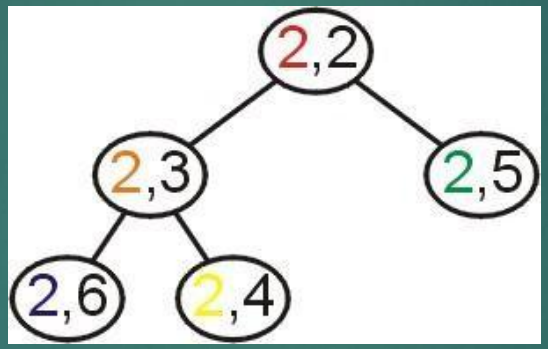
Popped: 2

- First, (2,1) < (2, 2) and (2, 3) < (2, 4)



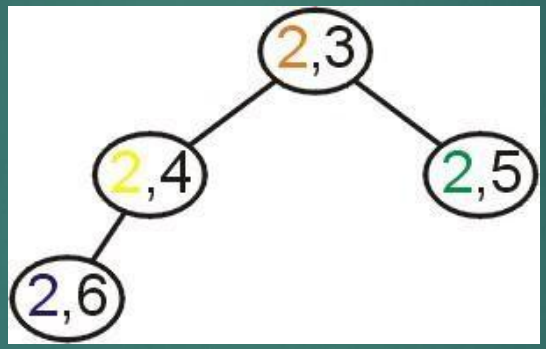
Priority Queues

Removing the objects would be in the following order:



Priority Queues

Removing the objects would be in the following order:



Priority Queues

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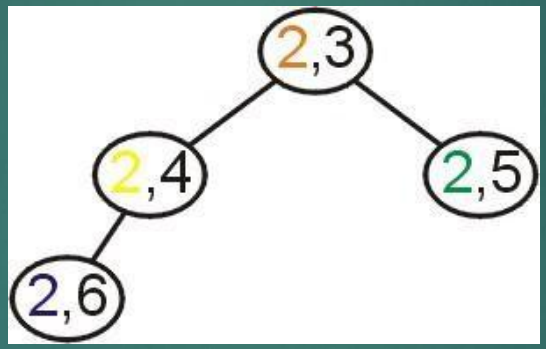

Priority Queues

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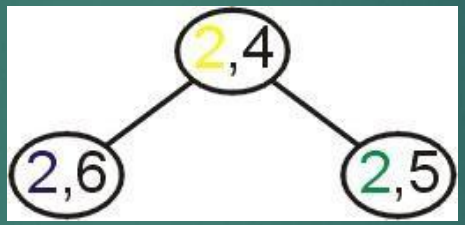
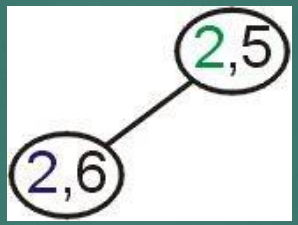


Priority Queues



Summary

In this talk, we have:

- Discussed binary heaps
- Looked at an implementation using arrays
- Discussed implementing priority queues using binary heaps
- Discussed Heap Sort
- The use of a lexicographical ordering